

Algorithms

Lecture #18

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Edit Distance

Edit Distance

DP

Formulation

Design & Analysis of Algorithms

Edit Distance

Edit Distance

S D I M D M
M A _ T H S
A _ R T _ S

Edit transcript: A string over the alphabet M, S, I, D that describes a transformation of one string into another. Example

$$\begin{array}{cccccc} S & D & I & M & D & M \\ 1+ & 1+ & 1+ & 0+ & 1+ & 0+ & = 4 \end{array}$$

Edit Distance

Edit Distance

- In general, it is not easy to determine the optimal edit distance.
- For example, the distance between ALGORITHM and ALTRUISTIC is at most 6.

A L G O R _ I _ T H M
A L _ T R U I S T I C

- Is this optimal?

Edit Distance: DP Algorithm

Edit Distance: DP Algorithm

Recursive definition:

- Suppose we have an m -character string A and an n -character string B .
- Define $E(i, j)$ to be the edit distance between the first i characters of A and the first j characters of B .

Edit Distance: DP Algorithm

Edit Distance: DP Algorithm

- $E(i, j)$:

$$\begin{array}{c} i \\ \underbrace{\text{A L G O R I T H M}} \\ \underbrace{\text{A L - T R U I S T I C}} \\ j \end{array}$$

- The edit distance between entire strings A and B is $E(m, n)$.

Edit Distance

Edit Distance

- The gap representation for the edit sequences has a crucial “optimal substructure” property.
- If we remove the last column, the remaining columns must represent the shortest edit sequence for the remaining substrings:

Edit Distance

Edit Distance

Edit distance = 6

A	L	G	O	R	_	I	_	T	H	M
A	L	_	T	R	U	I	S	T	I	C

Remove last column,

Edit distance = 5

A	L	G	O	R	_	I	_	T	H
A	L	_	T	R	U	I	S	T	I

Edit Distance

Edit Distance

There are a couple of obvious base cases:

- The only way to convert an empty string into a string of j characters is by doing j insertions. Thus

$$E(0, j) = j$$

- The only way to convert a string of i characters into the empty string is with i deletions:

$$E(i, 0) = i$$

Edit Distance

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Four possibilities for the last column in the shortest possible edit sequence:

Deletion: Last entry in bottom row is empty.

$i=3$										
A	L	G	O	R	I	T	H	M		
$j=2$										
A	L	_	T	R	U	I	S	T	I	C

In this case

$$E(i, j) = E(i - 1, j) + 1$$

Edit Distance

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Insertion: The last entry in the top row is empty.

$i=5$
A L G O R _ I T H M
A L _ T R U I S T I C
 $j=5$

In this case

$$E(i, j) = E(i, j - 1) + 1$$

Edit Distance

Edit Distance

Substitution: Both rows have characters in the last column.

$i=4$
A L G O R I T H M
A L - T R U I S T I C
 $j=3$

If the characters are different, then

$$E(i, j) = E(i - 1, j - 1) + 1$$

Edit Distance

Edit Distance

Substitution: Both rows have characters in the last column.

i=5
A L G O R I T H M
A L _ T R U I S T I C
j=4

If characters are same, no substitution is needed:

$$E(i, j) = E(i - 1, j - 1)$$

Edit Distance

Edit Distance

Thus the edit distance $E(i, j)$ is the smallest of the four possibilities:

$$E(i, j) = \min \begin{pmatrix} E(i-1, j) + 1 \\ E(i, j-1) + 1 \\ E(i-1, j-1) + 1 & \text{if } A[i] \neq B[j] \\ E(i-1, j-1) & \text{if } A[i] = B[j] \end{pmatrix}$$

Edit Distance

Edit Distance

Consider the example:

A	R	T	S	
M	A	T	H	S

The edit distance would be $E(4, 5)$.

Edit Distance

Edit Distance

If we recursion to compute, we will have

$$E(4,5) = \min \begin{pmatrix} E(3,5) + 1 \\ E(4,4) + 1 \\ E(3,4) + 1 & \text{if } A[4] \neq B[5] \\ E(3,4) & \text{if } A[4] = B[5] \end{pmatrix}$$

Recursion clearly leads to the same repetitive call pattern that we saw in Fibonacci sequence.

Edit Distance

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If we recursion to compute, we will have

$$E(4,5) = \min \begin{pmatrix} E(3,5) + 1 \\ E(4,4) + 1 \\ E(3,4) + 1 & \text{if } A[4] \neq B[5] \\ E(3,4) & \text{if } A[4] = B[5] \end{pmatrix}$$

To avoid this, we will use the DP approach.

Edit Distance

Edit Distance

If we recursion to compute, we will have

$$E(4,5) = \min \begin{pmatrix} E(3,5) + 1 \\ E(4,4) + 1 \\ E(3,4) + 1 & \text{if } A[4] \neq B[5] \\ E(3,4) & \text{if } A[4] = B[5] \end{pmatrix}$$

We will build the solution bottom-up.

Edit Distance

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Pattern of building $E(i, j)$

$E(i - 1, j - 1)$ $E(i - 1, j)$



$E(i, j - 1) \rightarrow E(i, j)$

Use the base case $E(0, j)$ to fill first row.

Use the base case $E(i, 0)$ to fill first column.

Fill the remaining E matrix row by row.

Edit Distance

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		A	R	T	S
	0	→ 1	→ 2	→ 3	→ 4
M	↓ 1	↘ 1	↘ 2	↘ 3	↘ 4
A	↓ 2	↘ 1	↘ 2	↘ 3	↘ 4
T	↓ 3	↓ 2	↘ 2	↘ 2	→ 3
H	↓ 4	↓ 3	↘ 3	↘ 3	↘ 3
S	↓ 5	↓ 4	↘ 4	↘ 4	↘ 3

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Edit Distance

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		A	R	T	S
	0	→ 1	→ 2	→ 3	→ 4
M	↓ 1	↘ 1	↘ 2	↘ 3	↘ 4
A	↓ 2	↘ 1	↘ 2	↘ 3	↘ 4
T	↓ 3	↓ 2	↘ 2	↘ 2	→ 3
H	↓ 4	↓ 3	↘ 3	↘ 3	↘ 3
S	↓ 5	↓ 4	↘ 4	↘ 4	↘ 3

Solution path 1:
 $1 + 0 + 1 + 1 + 0 = 3$
 D M S S M

 M A T H S
 – A R T S

Edit Distance

Edit Distance

		A	R	T	S
	0	→ 1	→ 2	→ 3	→ 4
M	↓ 1	↘ 1	↘ 2	↘ 3	↘ 4
A	↓ 2	↘ 1	↘ 2	↘ 3	↘ 4
T	↓ 3	↓ 2	↘ 2	↘ 2	→ 3
H	↓ 4	↓ 3	↘ 3	↘ 3	↘ 3
S	↓ 5	↓ 4	↘ 4	↘ 4	↘ 3

Solution path 2:
 $1 + 1 + 0 + 1 + 0 = 3$
 S S M D M

 M A T H S
 A R T _ S

Edit Distance

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		A	R	T	S
	0	→ 1	→ 2	→ 3	→ 4
M	↓ 1	↘ 1	↘ 2	↘ 3	↘ 4
A	↓ 2	↘ 1	↘ 2	↘ 3	↘ 4
T	↓ 3	↓ 2	↘ 2	↘ 2	→ 3
H	↓ 4	↓ 3	↘ 3	↘ 3	↘ 3
S	↓ 5	↓ 4	↘ 4	↘ 4	↘ 3

Solution path 3:

1+ 0+ 1+ 0+ 1+ 0=3

D M I M D M

M A _ T H S

_ A R T _ S

Edit Distance

Edit Distance

<i>S</i>	<i>D</i>	<i>I</i>	<i>M</i>	<i>D</i>	<i>M</i>
M	A	_	T	H	S
A	_	R	T	_	S

- **Edit transcript:** A string over the alphabet M, S, I, D that describes a transformation of one string into another Example.

<i>S</i>	<i>D</i>	<i>I</i>	<i>M</i>	<i>D</i>	<i>M</i>
1+	1+	1+	0+	1+	0+ = 4

Analysis: Edit Distance

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- There are $\Theta(n^2)$ entries in the matrix.
- Each entry $E(i, j)$ takes $\Theta(1)$ time to compute.
- The total running time is $\Theta(n^2)$.